Linear independence of the $c \to \infty$ half-space symmetric functions of Garbali–de Gier–Mead–Wheeler, and the resulting proof of their orthogonality conjecture

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Abstract

Garbali, de Gier, Mead and Wheeler [1] construct two families of symmetric functions G_ν, F_μ from the half-space stochastic six-vertex model and conjecture an orthogonality relation between them (their Conjecture conj:F-orth). They reduce the conjecture to a single unproved lemma, stating that it "remains conjectural as we have not established that the functions $\lim_{c\to\infty} G_\kappa(x_1,\dots,x_L)$ are linearly independent". We prove that lemma. The mechanism is a triangularity: the family of one-variable functions φ_m out of which G^∞_κ is built has a *nested zero locus* in the spectral variable x, so that evaluation at the points $x_i = y_{\kappa_i}^{-1}$ annihilates $G^\infty_{\kappa'}$ unless $\kappa' \preceq \kappa$ in an explicit partial order, while the diagonal value is a nonvanishing product computed in closed form. Because every principal down-set of $\preceq$ is *finite*, the same functional reduces an *infinite* relation to a finite one; this yields the topological form of independence which the authors' deduction actually requires, modulo one explicitly isolated interchange hypothesis. We also give a second, independent proof of the finite statement by iterated residues in x at the *poles* $x_i = (qy_{\kappa_i})^{-1}$, which is triangular for the *opposite* order. All structural lemmas are verified symbolically; the controls, including two that we show to be vacuous by construction and one planted error that fires, are reported in §7.

1 Introduction

1.1 What is proved

Let q, a and $y_1, y_2, \dots$ be indeterminates and set

$$\mathbb{K} := \mathbb{Q}\big(q, a, y_1, y_2, y_3, \dots\big).$$

Let $\mathcal{P}$ denote the set of finite subsets of $\mathbb{N} = \{1, 2, 3, \dots\}$, written in decreasing order, $\kappa = (\kappa_1 > \kappa_2 > \cdots > \kappa_d)$, $d = |\kappa|$; the empty set is allowed. For $\kappa \in \mathcal{P}$ and $L \geq 0$ let $G^\infty_\kappa(x_1,\dots,x_L) \in \mathbb{K}(x_1,\dots,x_L)$ be the $c \to \infty$ limit of the half-space partition function of [1], given explicitly in Definition 4 below.

Theorem 1 (Main theorem)**.** *Fix $L \geq 0$. The family*

$$\left\{ G_\kappa^\infty(x_1, \ldots, x_L) \;:\; \kappa \in \mathcal{P},\ |\kappa| \leq L \right\}$$

is linearly independent over $\mathbb{K}$.

This is verbatim the statement that [1, §6, after Conjecture `conj:F-orth`] records as missing. Theorem 1 is proved twice, by two different mechanisms (§4 and §5). The first mechanism gives more:

Theorem 2 (Topological form)**.** *Fix $L \geq 0$, let $S \subseteq \{\kappa \in \mathcal{P} : |\kappa| \leq L\}$ be an arbitrary — possibly infinite — set, and let $(c_\kappa)_{\kappa \in S}$ be elements of $\mathbb{K}$. Suppose $\sum_{\kappa \in S} c_\kappa G_\kappa^\infty = 0$ in a sense permitting termwise application of the functionals Φ_{κ^0} of Definition 13 (Hypothesis 15). Then $c_\kappa = 0$ for every $\kappa \in S$.*

Together with the authors' own reduction this gives their conjecture; see §6 for the precise statement and for the one hypothesis that remains.

1.2 A remark on priority

The authors of [1] write "we plan to return to its proof in a later text". They have therefore claimed this problem. The present note is offered as an independent proof of a lemma they state as open in a published paper; nothing here is intended to pre-empt their own treatment, which will presumably prove more.

2 Setup

2.1 The objects, measured off the source

We use [1] in the form of its LaTeX source `Half_line_ASEP.tex` (255,580 bytes) and compiled text layer (123,621 bytes); line numbers below refer to the source. The boundary weight function is

$$h(x) = \frac{ac(1-x^2)}{(a-x)(c-x)} \qquad (\texttt{h-intro}, \text{ line } 279), \tag{1}$$

and letting $c \to \infty$,

$$h^\infty(x) = \frac{a(1-x^2)}{a-x}, \qquad \zeta(x) := 1 - h^\infty(x) = \frac{(a-x) - a(1-x^2)}{a-x} = \frac{x(1-ax)}{x-a}. \tag{2}$$

The second equality reproduces the printed limit $\lim_{c\to\infty} Z_L(x_1, \ldots, x_L) = \prod_{i=1}^{L} x_i(1-ax_i)/(x_i - a)$ of [1, line 620] of the triangular partition function Z_L, which is our first control (§7).

Definition 3. For $m \geq 1$ put

$$\varphi_m(x) \;:=\; \frac{(1-q)\,x\,y_m}{1-q\,x\,y_m} \prod_{j=1}^{m-1} \frac{1-x\,y_j}{1-q\,x\,y_j} \;\in\; \mathbb{K}(x). \tag{3}$$

Definition 4. For $\kappa = (\kappa_1 > \cdots > \kappa_d) \in \mathcal{P}$ and $L \geq 0$, set

$$G_\kappa^\infty(x_1,\dots,x_L) := \sum_{\substack{K\subseteq[L]\\|K|=d}} \prod_{j\in\bar K}\zeta(x_j) \prod_{i\in K} h^\infty(x_i) \prod_{i\in K}\prod_{j\in\bar K}\left[\frac{x_j-qx_i}{x_j-x_i}\cdot\frac{1-x_ix_j}{1-qx_ix_j}\right] \prod_{\substack{i<j\\ i,j\in K}}\frac{1-x_ix_j}{1-qx_ix_j}$$
$$\times \sum_{\sigma\in S_d}\prod_{1\leq i<j\leq d}\frac{x_{k_{\sigma(j)}}-q\,x_{k_{\sigma(i)}}}{x_{k_{\sigma(j)}}-x_{k_{\sigma(i)}}}\prod_{i=1}^{d}\varphi_{\kappa_i}\big(x_{k_{\sigma(i)}}\big), \quad (4)$$

where $K=\{k_1<\cdots<k_d\}$ and $\bar K=[L]\setminus K$.

Equation (4) is the $c\to\infty$ specialisation of [1, Theorem `G nu thm`, line 3386]: there the first factor is $Z_{L-d}(x_{\bar K})$, which by (2) becomes $\prod_{j\in\bar K}\zeta(x_j)$. For $L<d$ the subset sum in (4) is *empty*, so $G_\kappa^\infty=0$; see Lemma 10.

Remark 5 (a typo in the source). The inner product in (3) is printed correctly as $\prod_{j=1}^{\nu_i-1}$ in `G nu sum expression thm eq` (line 3386), but as $\prod_{j-1}^{\nu_i-1}$ in the three later displays `eq: G nu integral forumla thm`, `...-cinf` and `F-int-orthog-2`. We read all four as $\prod_{j=1}^{\nu_i-1}$. This is flagged rather than silently corrected; it is a typesetting slip with no mathematical content, and the agreement of the subset formula with the integral formula (Control B, §7) confirms the reading.

2.2 The quantifier census

"Linearly independent" carries four quantifiers, and the first of them determines the method.

(1) Over which field? The relation whose coefficients are to be stripped is [1, `eq: G nu integral forumla thm-cinf2`, line 3857],

$$G_\nu^\infty(x_1,\dots,x_L) \;=\; \sum_\kappa A_{\nu,\kappa}\,G_\kappa^\infty(x_1,\dots,x_L), \quad (5)$$

where $A_{\nu,\kappa}$ is the n-fold contour integral printed there, $n=|\nu|$. *The coefficient $A_{\nu,\kappa}$ depends on the y-alphabet*: explicitly through the factors $y_{\nu_i}/(1-qw_iy_{\nu_i})\prod_{j<\nu_i}(1-w_iy_j)/(1-qw_iy_j)$, and again through $\lim_{c\to\infty}F_\kappa$. Hence the field must contain the y_j, and it is $\mathbb{K}$, with the x_i as the variables.

This decides the method. The obvious separating device — the y-support filtration, namely that G_κ^∞ involves y_j only for $j\leq\max\kappa$, so that a residue in $y_{\max\kappa}$ isolates the top level — is *unavailable*: residues in the y_j are not $\mathbb{K}$-linear, because the coefficients c_κ are themselves y-dependent. Every functional used below acts in the x-variables only, with the y_j as scalars.

(2) Which index set? κ runs over *all* finite subsets of $\mathbb{N}$: "the sum on the left is over all finite subsets $\kappa\subset\mathbb{N}$" [1, line 443]. In `conj:F-orth` the cardinality is further restricted to $|\kappa|\leq n$. No G_κ^∞ with $|\kappa|\leq L$ vanishes identically (Theorem 14 gives $\Phi_\kappa(G_\kappa^\infty)\neq 0$), so the index set needs no restriction beyond $|\kappa|\leq L$, which is forced by Lemma 10.

(3) At which L? `conj:F-orth` contains no L: its left-hand side is an n-fold integral in $w_1,\dots,w_n$ alone. L enters only through (5), which holds for every $L\geq n$. So independence at a *single* $L\geq n$ suffices. We prove it for every $L\geq 0$.

(4) Finite or infinite? The family and the relation (5) are both infinite. Theorem 1 is the finite statement (every finite subfamily independent), which is literally what [1] asks for; Theorem 2 is the infinite one, which is what (5) needs. The distinction is recorded here so that it cannot be elided later.

3 The order, and two structural lemmas

Definition 6. For $\kappa', \kappa \in \mathcal{P}$ write

$$\kappa' \preceq \kappa \qquad :\Longleftrightarrow \qquad |\kappa'| \le |\kappa| \quad \text{and} \quad \kappa'_j \le \kappa_j \quad (1 \le j \le |\kappa'|).$$

Lemma 7. *$\preceq$ is a partial order on $\mathcal{P}$ with least element $\varnothing$, and every principal down-set $\mathcal{P}_{\preceq\kappa} := \{\kappa' \in \mathcal{P} : \kappa' \preceq \kappa\}$ is* finite. *Consequently $\preceq$ is well-founded.*

Proof. Reflexivity and transitivity are immediate. If $\kappa' \preceq \kappa$ and $\kappa \preceq \kappa'$ then $|\kappa| = |\kappa'|$ and $\kappa'_j \le \kappa_j \le \kappa'_j$ for all j, so $\kappa = \kappa'$. If $\kappa' \preceq \kappa$ then $\kappa'_1 \le \kappa_1$, and since κ' is decreasing every element of κ' lies in $[\kappa_1]$; together with $|\kappa'| \le |\kappa|$ this leaves finitely many possibilities. A partial order all of whose principal down-sets are finite is well-founded. □

The finiteness of down-sets is the feature that makes Theorem 2 possible: it turns an infinite relation into a finite one in one stroke.

Lemma 8 (matching)**.** *Let $\kappa' = (\kappa'_1 > \cdots > \kappa'_{n'})$ and let $c_1, \dots, c_N \in \mathbb{N}$ with decreasing rearrangement $c_{[1]} \ge c_{[2]} \ge \cdots \ge c_{[N]}$. There is an injection $\iota : [n'] \to [N]$ with $\kappa'_j \le c_{\iota(j)}$ for all j if and only if $n' \le N$ and $\kappa'_j \le c_{[j]}$ for all $j \le n'$.*

Proof. ($\Leftarrow$) Let $\iota(j)$ be the index of the j-th largest entry (ties broken arbitrarily, consistently); this is injective and $c_{\iota(j)} = c_{[j]} \ge \kappa'_j$.

($\Rightarrow$) Let $e_1 \ge \cdots \ge e_{n'}$ be the decreasing rearrangement of $(c_{\iota(j)})_j$. Since ι is injective these are n' distinct entries of the list c, so the j-th largest among them is at most the j-th largest of the whole list: $e_j \le c_{[j]}$. Also, pointwise domination is preserved by decreasing rearrangement: from $\kappa'_j \le c_{\iota(j)}$ for all j we get $\kappa'_{[j]} \le e_j$, and κ' is already decreasing, so $\kappa'_j \le e_j \le c_{[j]}$. □

Lemma 9 (zeros and poles of φ_m)**.** *Fix $m, m' \ge 1$. In $\mathbb{K}(x)$:*

(i) φ_m has simple poles exactly at $x = (qy_j)^{-1}$ for $1 \le j \le m$, and $\varphi_m(0) = 0$. In particular the pole sets are nested, $\mathrm{Pol}(\varphi_1) \subset \mathrm{Pol}(\varphi_2) \subset \cdots$, *with $(qy_m)^{-1}$ a pole of $\varphi_{m'}$ if and only if $m' \ge m$.*

(ii) φ_m is regular at $x = y_{m'}^{-1}$, and

$$\varphi_m\big(y_{m'}^{-1}\big) = 0 \iff m > m'.$$

For $m \le m'$ one has $\varphi_m(y_{m'}^{-1}) = \dfrac{(1-q)y_m/y_{m'}}{1 - qy_m/y_{m'}} \prod_{j=1}^{m-1} \dfrac{1 - y_j/y_{m'}}{1 - qy_j/y_{m'}} \neq 0$; in particular

$$\varphi_m\big(y_m^{-1}\big) = \prod_{j=1}^{m-1} \frac{1 - y_j/y_m}{1 - qy_j/y_m} \;\neq\; 0. \tag{6}$$

(iii) $\operatorname{Res}_{x=(qy_m)^{-1}} \varphi_{m'} \neq 0$ *if and only if* $m' \geq m$.

Proof. (i) The denominators of (3) are $1 - qxy_j$, $1 \leq j \leq m$, which vanish exactly at $x = (qy_j)^{-1}$; these points are pairwise distinct and distinct from the zeros $x = y_j^{-1}$ $(j < m)$ of the numerator because the y_j are indeterminates and $q \neq 1$. The factor $(1-q)xy_m$ gives $\varphi_m(0) = 0$.

(ii) $y_{m'}^{-1} = (qy_j)^{-1}$ would force $y_j = q^{-1}y_{m'}$, impossible for indeterminates; so φ_m is regular there, and its value is obtained by substitution. If $m > m'$ then $m' \leq m-1$, so the index $j = m'$ occurs in $\prod_{j=1}^{m-1}$ and contributes $1 - y_{m'}/y_{m'} = 0$. If $m \leq m'$ then every $j \leq m-1$ satisfies $j < m \leq m'$, hence $j \neq m'$ and $1 - y_j/y_{m'} \neq 0$; also $1 - qy_m/y_{m'} \neq 0$ and $y_m/y_{m'} \neq 0$. Setting $m' = m$ gives (6), the first factor being $(1-q) \cdot 1/(1-q) = 1$.

(iii) By (i) the point $(qy_m)^{-1}$ is a pole of $\varphi_{m'}$ iff $m \leq m'$, and all poles are simple, so the residue is nonzero exactly then. □

Lemma 10 (vanishing)**.** $G_\kappa^\infty(x_1, \ldots, x_L) = 0$ *whenever* $|\kappa| > L$.

Proof. There is no $K \subseteq [L]$ with $|K| = |\kappa| > L$, so the sum (4) is empty. □

Remark 11 (why this is the right answer, not an artefact). [1, Theorem `G nu thm`] is stated under the hypothesis $L \geq n$, so Lemma 10 is formally a statement about the right-hand side of (4) only. It is however the correct value of the partition function. By [1, `G partition function defn A eq`, line 2200], $G_{\nu/\mu} = \langle \mu | A(x_1) \cdots A(x_L) | \nu \rangle$, where each double-row operator $A(x)$ contains exactly one boundary vertex; the bulk six-vertex weights conserve the number of paths at every vertex, and the external horizontal edges of a double row carry no path. Hence the occupation number changes by at most the amount a single boundary vertex can create, and in the $c \to \infty$ limit "the boundary vertices then only have the option to inject (but never eject) paths" [1, line 3076]: the number rises by at most one per double row. Starting from $\mu = \varnothing$ and applying L double rows gives $|\nu| \leq L$. So (4), read literally at $L < |\kappa|$, returns the correct value 0.

Lemma 12 (stability in L)**.** *For every* $\kappa \in \mathcal{P}$ *and every* $L \geq 1$*, the rational function* $G_\kappa^\infty(x_1, \ldots, x_L)$ *is regular at* $x_L = 1$ *and*

$$G_\kappa^\infty(x_1, \ldots, x_{L-1}, 1) \;=\; G_\kappa^\infty(x_1, \ldots, x_{L-1}).$$

Proof. Write $d = |\kappa|$ and split the sum (4) according to whether $L \in K$.

Every factor of (4) is regular at $x_L = 1$ when $x_1, \ldots, x_{L-1}$ are indeterminates: the possible poles are at $x_L \in \{a,\ x_i,\ (qx_i)^{-1},\ (qy_j)^{-1}\}$, none of which equals 1 in $\mathbb{K}(x_1, \ldots, x_{L-1})$.

If $L \in K$, the term contains the factor $h^\infty(x_L) = h^\infty(1) = a(1-1)/(a-1) = 0$ by (2), and all remaining factors of that term are finite at $x_L = 1$ by the previous paragraph; so the term vanishes.

If $L \notin K$, then $L \in \bar{K}$ and the entire x_L-dependence of the term is

$$\zeta(x_L) \prod_{i \in K} \left[\frac{x_L - qx_i}{x_L - x_i} \cdot \frac{1 - x_i x_L}{1 - qx_i x_L} \right].$$

At $x_L = 1$ the first factor is $\zeta(1) = 1 \cdot (1-a)/(1-a) = 1$, and each bracket is

$$\frac{1 - qx_i}{1 - x_i} \cdot \frac{1 - x_i}{1 - qx_i} = 1.$$

So the term equals the term of (4) for the alphabet $(x_1, \ldots, x_{L-1})$ with the same K. As K ranges over the d-subsets of $[L]$ not containing L it ranges exactly over the d-subsets of $[L-1]$, and the claim follows. (If $d = L$ the only K is $[L] \ni L$, and both sides are 0, consistently with Lemma 10.) □

4 The evaluation functional and the main theorem

Definition 13. Let $\kappa \in \mathcal{P}$, $d = |\kappa|$, and put

$$\xi_i \;:=\; y_{\kappa_i}^{-1} \qquad (1 \le i \le d).$$

For $L \ge d$ define Φ_κ on those $F \in \mathbb{K}(x_1, \dots, x_L)$ for which the following successive specialisations are defined:

$$\Phi_\kappa(F) \;:=\; \Big[\cdots \big[F\big|_{x_L=1} \big] \big|_{x_{L-1}=1} \cdots \Big] \Big|_{x_{d+1}=1} \;\text{ evaluated at }\; (x_1, \dots, x_d) = (\xi_1, \dots, \xi_d).$$

The substitutions are performed one variable at a time, each with the remaining variables still indeterminate; this is why no two variables are ever set equal, and no spurious pole at $x_i = x_j$ is created. Φ_κ is $\mathbb{K}$-linear on its domain of definition. By Lemma 12,

$$\Phi_\kappa\big(G^\infty_{\kappa'}(x_1, \dots, x_L)\big) \;=\; G^\infty_{\kappa'}(\xi_1, \dots, \xi_d) \qquad \text{for all } \kappa' \in \mathcal{P},\; L \ge d. \tag{7}$$

Theorem 14 (triangularity)**.** *Let $\kappa \in \mathcal{P}$, $d = |\kappa|$, $\xi_i = y_{\kappa_i}^{-1}$, and let $L \ge d$. Then for every $\kappa' \in \mathcal{P}$ the value $\Phi_\kappa(G^\infty_{\kappa'})$ is defined, and*

$$\Phi_\kappa\big(G^\infty_{\kappa'}\big) = 0 \qquad \textit{unless} \qquad \kappa' \preceq \kappa.$$

Moreover

$$\Phi_\kappa\big(G^\infty_{\kappa}\big) \;=\; D_\kappa \;:=\; \prod_{i=1}^{d} h^\infty(\xi_i) \prod_{1 \le i < j \le d} \left[\frac{1 - \xi_i\xi_j}{1 - q\xi_i\xi_j} \cdot \frac{\xi_j - q\xi_i}{\xi_j - \xi_i} \right] \prod_{i=1}^{d} \prod_{j=1}^{\kappa_i - 1} \frac{1 - y_j/y_{\kappa_i}}{1 - q\, y_j/y_{\kappa_i}} \;\neq\; 0. \tag{8}$$

Proof. By (7) it suffices to study $G^\infty_{\kappa'}(\xi_1, \dots, \xi_d)$. Put $n' = |\kappa'|$. If $n' > d$ then $G^\infty_{\kappa'}(\xi_1, \dots, \xi_d) = 0$ by Lemma 10, and also $\kappa' \not\preceq \kappa$; so assume $n' \le d$.

Regularity and nonvanishing of the prefactors. The $\xi_i = y_{\kappa_i}^{-1}$ are pairwise distinct, and in $\mathbb{K}$ one has $\xi_i \notin \{0, a, \pm 1, a^{-1}\}$, $\xi_i\xi_j \notin \{1, q^{-1}\}$, $\xi_j \neq q\xi_i$, and $\xi_i \neq (qy_j)^{-1}$ for all j, because the y_j, q, a are algebraically independent. Hence for every $K \subseteq [d]$ with $|K| = n'$ and every $\sigma \in S_{n'}$, all the factors

$$\zeta(\xi_j)\ (j \in \bar{K}), \quad h^\infty(\xi_i)\ (i \in K), \quad \frac{\xi_j - q\xi_i}{\xi_j - \xi_i}, \quad \frac{1 - \xi_i\xi_j}{1 - q\xi_i\xi_j}, \quad \frac{\xi_{k_{\sigma(j)}} - q\xi_{k_{\sigma(i)}}}{\xi_{k_{\sigma(j)}} - \xi_{k_{\sigma(i)}}}$$

are defined and nonzero, and by Lemma 9(ii) each $\varphi_{\kappa'_i}(\xi_\cdot)$ is defined. Therefore the (K, σ) term of (4) vanishes *if and only if* one of its factors $\varphi_{\kappa'_i}(\xi_{k_{\sigma(i)}})$ vanishes.

The criterion. By Lemma 9(ii), $\varphi_{\kappa'_i}(\xi_{k_{\sigma(i)}}) = \varphi_{\kappa'_i}(y^{-1}_{\kappa_{k_{\sigma(i)}}}) = 0$ if and only if $\kappa'_i > \kappa_{k_{\sigma(i)}}$. Hence the (K, σ) term is nonzero if and only if

$$\kappa'_i \;\le\; \kappa_{\iota(i)} \quad (1 \le i \le n'), \qquad \text{where } \iota(i) := k_{\sigma(i)}, \tag{9}$$

and $\iota : [n'] \to [d]$ is an injection. Conversely every injection $\iota : [n'] \to [d]$ arises from exactly one pair (K, σ), namely $K = \iota([n'])$ and σ the induced permutation. Applying Lemma 8 with caps $c_t = \kappa_t$ $(t \in [d])$, which are already in decreasing order so that $c_{[j]} = \kappa_j$, we conclude:

$$\text{some } (K, \sigma) \text{ term is nonzero} \iff n' \le d \text{ and } \kappa'_j \le \kappa_j\ (j \le n') \iff \kappa' \preceq \kappa.$$

So if $\kappa' \not\preceq \kappa$ then *every* term of (4) vanishes at $(\xi_1, \dots, \xi_d)$ and $\Phi_\kappa(G^\infty_{\kappa'}) = 0$. (No cancellation argument is needed: the terms vanish individually.)

The diagonal. Let $\kappa' = \kappa$, so $n' = d$ and necessarily $K = [d]$, $k_t = t$. The criterion (9) reads $\kappa_i \le \kappa_{\sigma(i)}$ for all i; since κ is strictly decreasing this is equivalent to $\sigma(i) \le i$ for all i, which forces $\sigma = \mathrm{id}$. Hence exactly one term survives. Its value is, using $\bar{K} = \varnothing$ and (6),

$$\prod_{i=1}^{d} h^\infty(\xi_i) \prod_{1 \le i < j \le d} \frac{1 - \xi_i \xi_j}{1 - q\xi_i \xi_j} \prod_{1 \le i < j \le d} \frac{\xi_j - q\xi_i}{\xi_j - \xi_i} \prod_{i=1}^{d} \varphi_{\kappa_i}\big(y_{\kappa_i}^{-1}\big) \;=\; D_\kappa,$$

a product of finitely many nonzero elements of $\mathbb{K}$, hence nonzero. For $d = 0$ the empty product gives $\Phi_\varnothing(G^\infty_\varnothing) = D_\varnothing = 1$. □

Proof of Theorem 1. Suppose $\sum_{\kappa \in S} c_\kappa G^\infty_\kappa(x_1, \dots, x_L) = 0$ with S finite, nonempty, all $c_\kappa \in \mathbb{K}\backslash\{0\}$, and $|\kappa| \le L$ for $\kappa \in S$. Since S is finite and nonempty it contains a $\preceq$-minimal element κ^0. Apply the $\mathbb{K}$-linear functional Φ_{κ^0}. By Theorem 14 only those $\kappa \in S$ with $\kappa \preceq \kappa^0$ survive, and by minimality the only one is κ^0 itself. Hence $c_{\kappa^0} D_{\kappa^0} = 0$ with $D_{\kappa^0} \neq 0$, so $c_{\kappa^0} = 0$ — a contradiction. □

Hypothesis 15 (termwise specialisation). In the setting of Theorem 2, for each $\kappa^0 \in \mathcal{P}$ with $|\kappa^0| \le L$ the identity $\sum_{\kappa \in S} c_\kappa G^\infty_\kappa = 0$ remains valid after applying Φ_{κ^0} termwise, i.e. $\sum_{\kappa \in S} c_\kappa \Phi_{\kappa^0}(G^\infty_\kappa) = 0$.

Proof of Theorem 2. We argue by well-founded induction on $\preceq$, legitimate by Lemma 7. Let $\kappa^0 \in \mathcal{P}$ and assume $c_\kappa = 0$ for every $\kappa \in S$ with $\kappa \prec \kappa^0$. Apply Φ_{κ^0} to the relation (Hypothesis 15). By Theorem 14 every term with $\kappa \not\preceq \kappa^0$ drops out, so

$$\sum_{\kappa \in S \cap \mathcal{P}_{\preceq \kappa^0}} c_\kappa \, \Phi_{\kappa^0}(G^\infty_\kappa) \;=\; 0,$$

and this is a *finite* sum because $\mathcal{P}_{\preceq \kappa^0}$ is finite (Lemma 7). By the inductive hypothesis all its terms with $\kappa \neq \kappa^0$ vanish, leaving $c_{\kappa^0} D_{\kappa^0} = 0$, whence $c_{\kappa^0} = 0$ (taking $c_{\kappa^0} := 0$ if $\kappa^0 \notin S$). □

Remark 16 (what Hypothesis 15 is and is not). Hypothesis 15 is an *interchange* assumption — that a convergent sum may be specialised termwise — not a disguised independence assumption. It is automatic when S is finite, which is why Theorem 1 needs no hypothesis. For infinite S it would follow from locally uniform convergence of the κ-sum on a neighbourhood, in the x-variables, of each configuration $(\xi_1, \dots, \xi_d, 1, \dots, 1)$. We do not verify that analytic point and do not claim it; see §8. Two features of the authors' own convergence regime make it plausible: that regime requires the points y_k^{-1} to be "arbitrarily close to the points x_i" [1, line 3869] — which is exactly the configuration $x_i = y_{\kappa_i}^{-1}$ — and they note that their contour conditions "allow us to choose elements of our alphabet, x_i, to be arbitrarily close to 1" [1, line 3737].

5 A second, independent mechanism: residues at the poles

The functional of §4 uses the *zeros* of φ_m. The *poles* give a second separating family, triangular for the *opposite* order. We record it because a corroboration obtained by a different mechanism is worth more than a second reading of one mechanism; it also reproves Theorem 1 with no reference to Lemma 12 or Lemma 10.

Definition 17. For $\kappa \in \mathcal{P}$, $d = |\kappa|$, and $L \geq d$ let

$$R_\kappa \;:=\; \operatorname{Res}_{x_d=(qy_{\kappa_d})^{-1}} \;\cdots\; \operatorname{Res}_{x_1=(qy_{\kappa_1})^{-1}},$$

an iterated residue leaving $x_{d+1}, \ldots, x_L$ free. Write $\kappa' \succeq \kappa$ for $|\kappa'| \geq |\kappa|$ and $\kappa'_j \geq \kappa_j$ $(j \leq |\kappa|)$.

Proposition 18. *Let $L \geq d = |\kappa|$. For every $\kappa' \in \mathcal{P}$ with $|\kappa'| \leq L$,*

$$R_\kappa\big(G^\infty_{\kappa'}\big) = 0 \qquad \textit{unless} \qquad \kappa' \succeq \kappa,$$

and $R_\kappa(G^\infty_\kappa) \neq 0$; explicitly, with $\xi_i = (qy_{\kappa_i})^{-1}$,

$$R_\kappa(G^\infty_\kappa) = \prod_{j=d+1}^{L} \zeta(x_j) \prod_{i=1}^{d} h^\infty(\xi_i) \prod_{i=1}^{d} \prod_{j=d+1}^{L} \left[\frac{x_j - q\xi_i}{x_j - \xi_i} \frac{1 - \xi_i x_j}{1 - q\xi_i x_j}\right] \prod_{1 \leq i < j \leq d} \left[\frac{1 - \xi_i\xi_j}{1 - q\xi_i\xi_j} \frac{\xi_j - q\xi_i}{\xi_j - \xi_i}\right] \prod_{i=1}^{d} \rho_{\kappa_i},$$

where $\rho_m := \operatorname{Res}_{x=(qy_m)^{-1}} \varphi_m \neq 0$. Consequently the family $\{G^\infty_\kappa : |\kappa| \leq L\}$ is $\mathbb{K}$-linearly independent.

Proof. Among the factors of (4), the only ones with a pole on the locus $x_i = (qy_m)^{-1}$ are the φ's: ζ and h^∞ have their poles at $x = a$, and the remaining factors at $x_i = x_j$, $x_i x_j = q^{-1}$, which are distinct from $(qy_m)^{-1}$ in $\mathbb{K}$. Residues are $\mathbb{K}$-linear, so we may work term by term; each term has at most a simple pole in each x_i there.

Fix κ' with $n' = |\kappa'|$ and a term (K, σ). In that term the variable x_t carries the factor $\varphi_{\kappa'_{\sigma^{-1}(s)}}$ where $t = k_s$, if $t \in K$, and carries no φ at all if $t \notin K$. Taking the residue in $x_1, \ldots, x_d$ therefore gives 0 unless $[d] \subseteq K$ — in particular unless $n' \geq d$ — and, since K is listed increasingly, $[d] \subseteq K$ forces $k_t = t$ for $t \leq d$. By Lemma 9(iii) the residue at $x_t = (qy_{\kappa_t})^{-1}$ of $\varphi_{\kappa'_{\sigma^{-1}(t)}}$ is nonzero iff $\kappa'_{\sigma^{-1}(t)} \geq \kappa_t$. Setting $\iota(t) := \sigma^{-1}(t)$, an injection $[d] \to [n']$, the term survives only if $\kappa'_{\iota(t)} \geq \kappa_t$ for $t \leq d$. Applying Lemma 8 to the reversed inequalities (equivalently, to the order-reversing relabelling) this is possible iff $\kappa'_t \geq \kappa_t$ for $t \leq d$, i.e. iff $\kappa' \succeq \kappa$.

For $\kappa' = \kappa$ we have $n' = d$ and $[d] \subseteq K$ with $|K| = d$ forces $K = [d]$; the condition $\kappa_{\sigma^{-1}(t)} \geq \kappa_t$ for all t means $\sigma^{-1}(t) \leq t$, i.e. $\sigma = \mathrm{id}$. One term survives and the displayed product is a product of nonzero elements of $\mathbb{K}(x_{d+1}, \ldots, x_L)$.

Independence: given a finite relation with nonempty support S, pick κ^0 $\succeq$-maximal in S; applying R_{κ^0} kills every other term and leaves $c_{\kappa^0} R_{\kappa^0}(G^\infty_{\kappa^0}) = 0$, so $c_{\kappa^0} = 0$. □

Remark 19. Proposition 18 does *not* give Theorem 2: $\succeq$ has infinite ascending chains, so a $\succeq$-maximal element of an infinite support need not exist. This asymmetry — the zero side is well-founded, the pole side is not — is the reason §4 is the primary argument and this section is corroboration.

6 Consequence: the orthogonality conjecture

We now record what the two theorems give for [1, Conjecture `conj:F-orth`]. The reduction is the authors'; we have read it (lines 3840–3907) and restate it so that the logical dependence is visible.

Fix a finite $\nu = (\nu_1 > \cdots > \nu_n)$ and $L \geq n$. Combining the $c \to \infty$ Cauchy identity `cauchy-cor-cinf` with the $c \to \infty$ integral formula `eq: G nu integral forumla thm-cinf`, the authors obtain `eq: G nu integral forumla thm-cinf2`, which is exactly (5) with

$$A_{\nu,\kappa} = \oint_{\mathcal{C}} \frac{dw_1}{2\pi i} \cdots \oint_{\mathcal{C}} \frac{dw_n}{2\pi i} \prod_{1 \leq i < j \leq n} \left[\frac{w_j - w_i}{q w_j - w_i} \frac{1 - q w_i w_j}{1 - w_i w_j} \right] \times \prod_{i=1}^{n} \left[\frac{w_i - a}{w_i(1 - a w_i)} \frac{1 - q w_i^2}{1 - w_i^2} \frac{y_{\nu_i}}{1 - q w_i y_{\nu_i}} \prod_{j=1}^{\nu_i - 1} \frac{1 - w_i y_j}{1 - q w_i y_j} \right] \lim_{c \to \infty} F_\kappa(w_1, \ldots, w_n), \quad (10)$$

the sum in (5) running over all finite $\kappa \subset \mathbb{N}$, and the contour $\mathcal{C}$ as in `conj:F-orth`. Rewriting (5) as $\sum_\kappa \big(A_{\nu,\kappa} - \delta_{\kappa,\nu}\big) G_\kappa^\infty = 0$, the conjecture $A_{\nu,\kappa} = \delta_{\kappa,\nu}$ is precisely the assertion that these coefficients vanish.

Corollary 20. *Let ν be a finite subset of $\mathbb{N}$, $n = |\nu|$, and suppose*

(a) the identity (5) holds at some $L \geq n$, with coefficients $A_{\nu,\kappa} \in \mathbb{K}$ as in (10), and

(b) Hypothesis 15 holds for that relation.

Then $A_{\nu,\kappa} = \delta_{\kappa,\nu}$ for all finite $\kappa \subset \mathbb{N}$ with $|\kappa| \leq L$; that is, `conj:F-orth` (5) of [1] holds. If moreover only finitely many $A_{\nu,\kappa}$ are nonzero, (b) is unnecessary and the conclusion follows from Theorem 1 alone.

Proof. Apply Theorem 2 with $c_\kappa = A_{\nu,\kappa} - \delta_{\kappa,\nu}$; terms with $|\kappa| > L$ have $G_\kappa^\infty = 0$ by Lemma 10 and are excluded from the conclusion. The final sentence is Theorem 1. □

Item (a) is the authors' own derivation and we do not reprove it; we have however checked that it is complete as a derivation, i.e. that the only step it leaves open is the independence of the G_κ^∞. The coefficients $A_{\nu,\kappa}$ lie in $\mathbb{K}$ because the integrand of (10) is a rational function of w and of q, a, Y and the integral is a finite sum of its residues.

7 Computational verification

All computations are exact, in pure Python with SymPy (SageMath is not available in this environment); code in `projects/code/Q473/`. Parameters q, a, y_j were specialised to distinct rationals and the x_i kept symbolic or specialised to rationals as indicated. Every residual below is an *exact* zero, not a numerical one.

Control 1 (non-vacuity of the engine). $G_\varnothing^\infty(x_1, \ldots, x_L)$ must equal the printed $\prod_i x_i(1 - a x_i)/(x_i - a)$ [1, line 620]. Residual 0 for $L = 1, 2, 3, 4$. **4/4.**

Control A (symmetry) — *vacuous by construction; reported as such.* G_κ^∞ is symmetric in $x_1, \ldots, x_L$, which is not manifest in (4). Verified for 6 pairs (κ, L). But the planted sign error of Control 3 *also* passes it: the inner sum $\sum_{\sigma \in S_d}$ is symmetric in the x's whatever its summand, so this control cannot see the plant. It is recorded as reading nothing.

Control B (two independently derived formulas). The subset formula (4) against the contour-integral formula `eq: G nu integral forumla thm-cinf` (line 3850), evaluated as a sum of residues at $w_i = x_{j_i}$ over injective tuples, per the authors' prescription at lines 3802 and 3821. Residual 0 at two independent random parameter points for $(\kappa, L) \in \{((1),1),((1),2),((2),2),((2,1),2),((3,1)$ **7/7, i.e. 14/14 evaluations.**

Control 3 (planted error) — *fires.* Flipping one sign in the S_d product $(x_{k_{\sigma(j)}} - q x_{k_{\sigma(i)}} \mapsto x_{k_{\sigma(j)}} + q x_{k_{\sigma(i)}})$ breaks Control B for all five cases with $|\kappa| \geq 2$ (residuals of order 1 to 10^3), and leaves the three cases with $|\kappa| \leq 1$ untouched — correctly, since the product over $i < j$ is then empty. Control 1 is likewise blind to the plant ($d = 0$), as was pre-registered.

Control C (stability, Lemma 12). $G^{\infty}_{\kappa}(x_1, \ldots, x_{L-1}, 1) - G^{\infty}_{\kappa}(x_1, \ldots, x_{L-1}) = 0$ symbolically, for $\kappa \in \{\varnothing, (1), (2), (2,1), (3,1), (3,2,1)\}$ and all admissible $L \leq 3$. **14/14.**

Control D (vanishing, Lemma 10). $G^{\infty}_{\kappa}(x_1, \ldots, x_L) = 0$ for $(\kappa, L) \in \{((2,1),1),((3,2,1),2),((3,2,1),1)\}$. **3/3.**

Main check: evaluation triangularity (Theorem 14). For all $\kappa, \kappa' \subseteq \{1,2,3,4\}$ with $|\kappa|, |\kappa'| \leq 3$, the predicted equivalence $\big(\Phi_{\kappa}(G^{\infty}_{\kappa'}) = 0\big) \Leftrightarrow \big(\kappa' \not\preceq \kappa\big)$ holds in **210/210** cells, comprising **101 zero** and **109 nonzero** cells — so the prediction was at risk in both directions. The diagonal was nonzero in **14/14** cases and agreed with the closed form (8) in **14/14**.

Main check: residue triangularity (Proposition 18). At $L = 2$, for all $\kappa, \kappa' \subseteq \{1,2,3\}$ with $|\kappa|, |\kappa'| \leq 2$: **42/42** cells match, **22 zero** and **20 nonzero**; diagonal nonzero **6/6**. At $L = 3$, for all $\kappa, \kappa' \subseteq \{1,2,3,4\}$ with $|\kappa|, |\kappa'| \leq 3$: **210/210** cells match, **115 zero** and **95 nonzero**; diagonal nonzero **14/14**. Note the zero/nonzero split differs from the evaluation table (115/95 against 101/109) over the same index range: the two functionals are triangular for *opposite* orders, so this is the expected signature and not a repeat reading of one engine.

Control 2 (planted dependency) — *fires correctly.* The matrix of the 11 functions G^{∞}_{κ}, $\kappa \subseteq \{1,2,3,4\}$, $|\kappa| \leq 2$, evaluated at 15 random rational points of (x_1, x_2) at $L = 2$, has rank 11 — full. Appending the column $G^{\infty}_{\kappa_1} + G^{\infty}_{\kappa_2}$ leaves the rank at 11, as it must; had it risen, the rank engine would have been broken.

Control 4 (two engines). The same rank computed exactly over $\mathbb{Q}$ and again modulo the Mersenne prime $p = 2^{31} - 1$: both give 11. The two agree, so neither is reading an artefact of the arithmetic.

The stratum, printed. Verified: $\max \kappa \leq 4$, $|\kappa| \leq 3$ for the evaluation triangularity; $\max \kappa \leq 3$, $|\kappa| \leq 2$, $L = 2$ for the residue triangularity; $L \leq 4$ for Control 1; $L \leq 3$ elsewhere. *Not* reached by computation: any κ with $\max \kappa \geq 5$ or $|\kappa| \geq 4$, and any $L \geq 4$ outside Control 1. These lie inside the stratum of Theorems 1–14, so they are covered by the proof and not by the computation; the proofs of Lemmas 7–12 and Theorem 14 are uniform in κ and L and use no case analysis that the tested range could hide.

8 Gaps and limits, stated precisely

1. **Theorem 1 has no gap.** It is purely algebraic, over $\mathbb{K} = \mathbb{Q}(q, a, Y)$ with the y_j indeterminate. If one wants it for specific complex parameters, the requirement is that the finitely many elements of $\mathbb{K}$ appearing in D_κ and in the genericity list in the proof of Theorem 14 be nonzero; these are explicit.

2. **Hypothesis 15 is the one open point** in the route to `conj:F-orth`. It is an interchange of a specialisation with an infinite sum (Remark 16). It is *not* needed for Theorem 1. We have not carried out the estimate that would place the configurations $(y_{\kappa_1}^{-1}, \dots, y_{\kappa_d}^{-1}, 1, \dots, 1)$ inside the authors' convergence domain `Cauchy identity condition-2` with locally uniform convergence, and we do not assert it.

3. **Remark 11 is our reading, not a printed statement.** The paper states $L \geq n$ as a hypothesis and (at $c \to \infty$) that boundary vertices only inject; the deduction that $G_\nu = 0$ for $|\nu| > L$ is ours. It is used only to interpret Lemma 10, which itself is immediate from (4) being an empty sum, so nothing in the proofs depends on Remark 11.

4. **The source typo** of Remark 5 ($\prod_{j-1}$ for $\prod_{j=1}$ in three displays) is read as intended. Control B confirms the reading.

5. **A control that read nothing, recorded as such.** Control A (symmetry in the x_i) passes both for the correct formula and for the planted sign error, because $\sum_{\sigma \in S_d}$ is symmetric in the x's whatever its summand. It is therefore vacuous by construction and contributes no evidence. We note it because it was the first control we ran and it looked informative: the plant is what revealed it.

6. **The rank arm at $L = 3$** was still computing when this note was closed; the reported rank evidence is at $L = 2$ only. The primary evidence is in any case the exact symbolic zero/nonzero pattern, not a rank.

7. **Priority.** The authors state they will return to this proof. See §1.2.

References

[1] A. Garbali, J. de Gier, W. Mead, M. Wheeler, *Symmetric functions from the six-vertex model in half-space*, `arXiv:2312.14348`. Source file `Half_line_ASEP.tex`; line numbers as cited.